

- 9 (a) The Principle of Superposition states that when two or more waves of the same kind overlap, the resultant displacement at any point at any instant B1
B1

is the vector sum of the displacements that the individual waves would have separately produced at that point and at that instant.

(b) $d \sin \theta = n\lambda$ M1

(i) $\left(\frac{10^{-3}}{550}\right) \sin 90 = n(644 \times 10^{-9})$ A1

$$n = 2.8$$

Highest order that can be seen is the 2nd order.

Hence total number of bright lines observed is 5. A1

(ii) 1. Since θ is greater, λ is also greater. B1

2. When n is larger, $\Delta\theta$ is larger, thus the greatest separation occurs in the second order. B1

(c) (i) The coherent waves from the laser meet at a point on the screen with a constant phase/path difference. B1

When waves meet in phase with phase difference of $n(2\pi \text{ rad})$ or path difference of $n\lambda$ where n is an integer, constructive interference occurs. B1

When the waves meet exactly out of phase (any equivalent explanation of minima e.g. $(n+\frac{1}{2})\lambda$ or $(n+\frac{1}{2}) \times 2\pi \text{ rad}$, destructive interference occurs. B1

(ii) 1.

$$\sin \theta = \frac{\lambda}{b} \quad \text{M1}$$

$$\sin \theta = (6.44 \times 10^{-7}) / (2.4 \times 10^{-5}) \quad \text{A1}$$

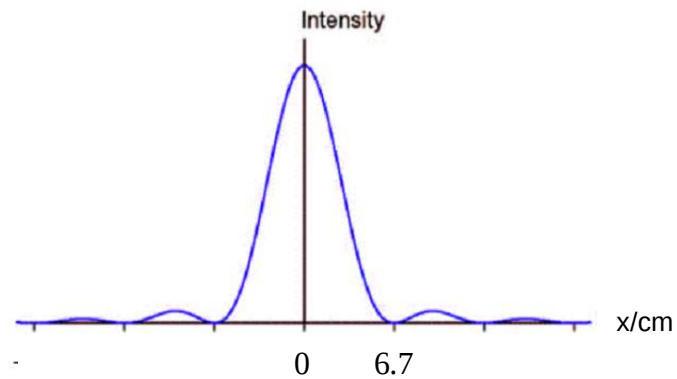
$$\theta = 1.54^\circ$$

$$\tan 1.54^\circ = O'Y / 2.5$$

$$O'Y = 0.067 \text{ m}$$

$$\text{Width} = 0.067 \times 2 = 0.134 \text{ m} \quad \text{A1}$$

2.



B1
B1
B1

Diagram should show 3 maximas, symmetrical about O
Correct shape, with decreasing intensity
Correct spacing labelled
Any missing- minus 1 mark

(ii) 3.

$$I \propto A^2$$

With 2 slits,

When Intensity = I , amplitude of maxima = $A + A$

With 1 slit,

Amplitude of maxima = A

$$I' = (A/2A)^2 \times I = \frac{1}{4} I$$

C1

A1

(iii) Rayleigh criterion stated.

B1

Yes as the interference patterns formed by the two sources are resolved when θ is equal or greater than 1.54° .

B1